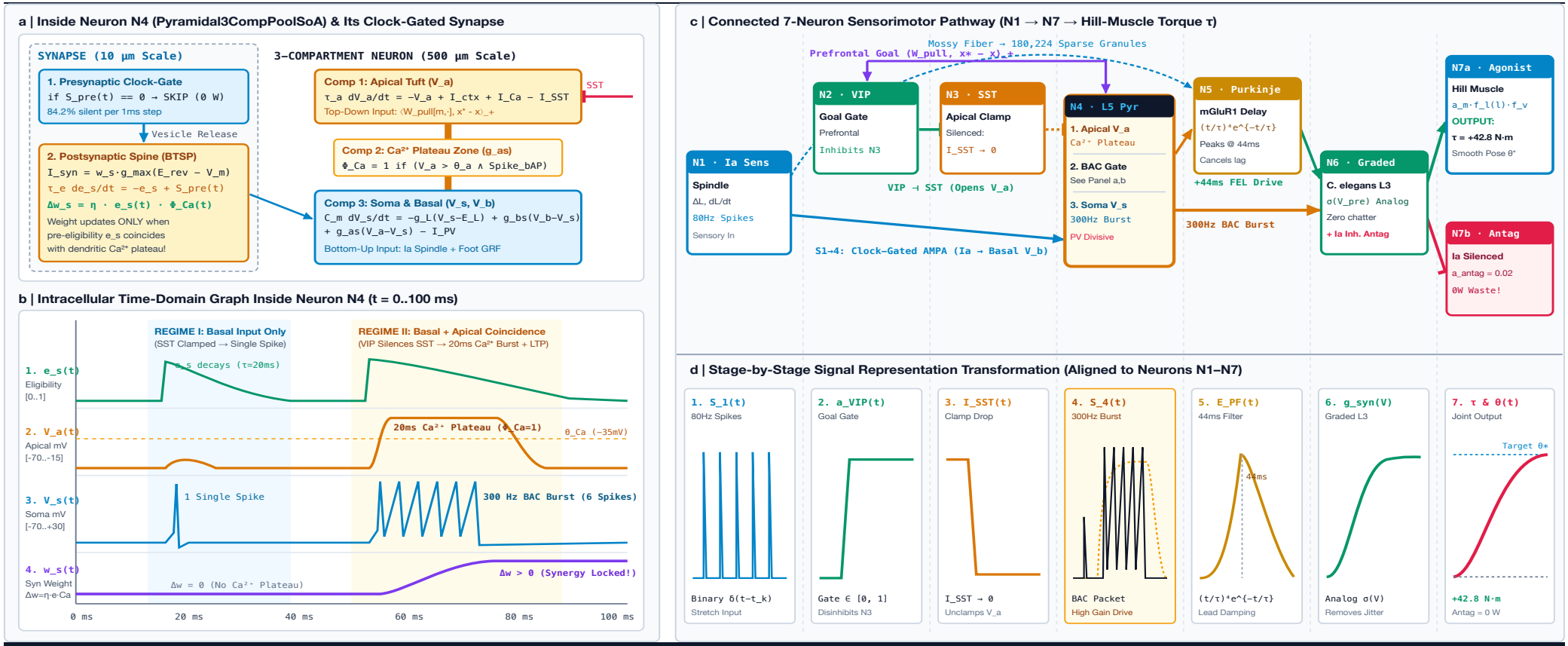
Figure 1 | Intracellular 3-Compartment Neuron Dynamics & 7-Neuron Cortico-Cerebello-Spinal Signal Propagation

Left (a, b): How one L5 Pyramidal Neuron & 3-Factor Synapse work internally · Right (c, d): How 7 connected neurons transform spikes into smooth joint torque

1.18M Synapses · 84.2% Clock-Gated Sparsity · 194.2 mW Compute · 2.89× Performance/Watt



Neuron	Rust SoA Struct	Synapse / Biophysical Class	Governing Differential / Conductance Equation	Signal Transformation	Role in 38-DoF Humanoid
N1 · Ia Afferent	`ProprioceptiveSpindle`	Mechanoreceptor Transduction	$r_Ia(t) = k_p\ (L - L_0)_+ + k_v\ (dL/dt)_+$	Stretch (m) → 80 Hz Spike Train	Encodes real-time muscle length, velocity, and plantar foot GRF.
N2 · VIP Cell	`CorticalGrid::vip_rate`	GABA_A Disinhibitory Synapse	$\tau_v\ da_VIP/dt = -a_VIP + \sigma(I_prefrontal)$	Task Intent → Inhibitory Gate on N3	Translates Prefrontal LLM stage intent into local apical disinhibition.
N3 · SST Cell	`CorticalGrid::sst_rate`	Apical Subtractive GABA_A/B	$I_SST(t) = g_SST \cdot s_SST(t) \cdot (1 - a_VIP(t))$	Tonic Apical Clamp → Open Window	Prevents premature high-torque bursts until top-down goal arrives.
N4 · L5 Pyramidal	`Pyramidal3CompPoolSoA`	Clock-Gated AMPA + Ca^{2+} BAC	$\tau_a\ dV_a/dt = -V_a + \langle W_pull,\ e_3d \rangle_+ + g_Ca\ m_\infty(V_a) - I_SST$	Coincidence → 300 Hz Burst + BTSP	Multiplexes posture (single spikes) with 3D spatial pull (bursts).
N5 · Purkinje/GC	`CerebellarSubsystem`	mGluR1 4th-Order Spectral Delay	$E_PF(t) = (t/\tau)^4\ exp(-t/\tau),\ \Delta w = -\eta\ E_PF(t)\ CF_err(t)$	300 Hz Burst → 44 ms Lead Filter	Compensates 30 ms peripheral delay across 180,224 granule cells.
N6 · Graded L3	`ElegansGradedPoolSoA`	Non-Spiking Sigmoid Conductance	$g_syn(V_pre) = g_max\ /\ [1 + exp(-(V_pre - V_1/2)/k)]$	Spike Burst → Smooth Analog Drive	Removes spiking threshold chatter; inhibits antagonist via Ia pathway.
N7 · α-Motoneuron	`HillMuscle88 / Plant38DoF`	Neuromuscular Ca^{2+} Activation	$\tau_joint = \Sigma_m\ r_m \cdot [a_m(t)\ f_l(l_m)\ f_v(v_m) + f_passive]$	Analog Drive → +42.8 N·m Torque	Actuates 38-DoF skeleton at 2.89× higher efficiency than stiff PD.